LAB 8: Server Configuration-DHCP, DNS, and Web Server in Cisco Packet Tracer

Objectives

1. To examine the working principle of the Dynamic Host Configuration Protocol (DHCP) and understand how it enables automatic IP address distribution across a network.

2. To set up a Domain Name System (DNS) server that resolves human-readable domain names into their corresponding IP addresses.

3. To install and configure a Web Server (HTTP) for hosting a webpage, and verify access from client PCs via a web browser.

Theory

1. DHCP (Dynamic Host Configuration Protocol)

DHCP is a network administration protocol that operates over UDP/IP networks.

Operation: Upon connecting to a network, a device sends a broadcast request, and the DHCP server responds by automatically allocating an IP address along with configuration details such as the subnet mask and default gateway. This removes the burden of manually assigning IP addresses to individual devices.

Application: Widely deployed in both corporate and residential networks to streamline IP management and eliminate address conflicts.

2. DNS (Domain Name System)

DNS is a distributed, hierarchical naming infrastructure used to identify computers, services, and resources connected to the Internet or a private network.

Operation: It converts user-friendly domain names (e.g., www.hcoe.com) into the corresponding numerical IP addresses (e.g., 192.168.1.2) that network devices rely on to communicate with each other.

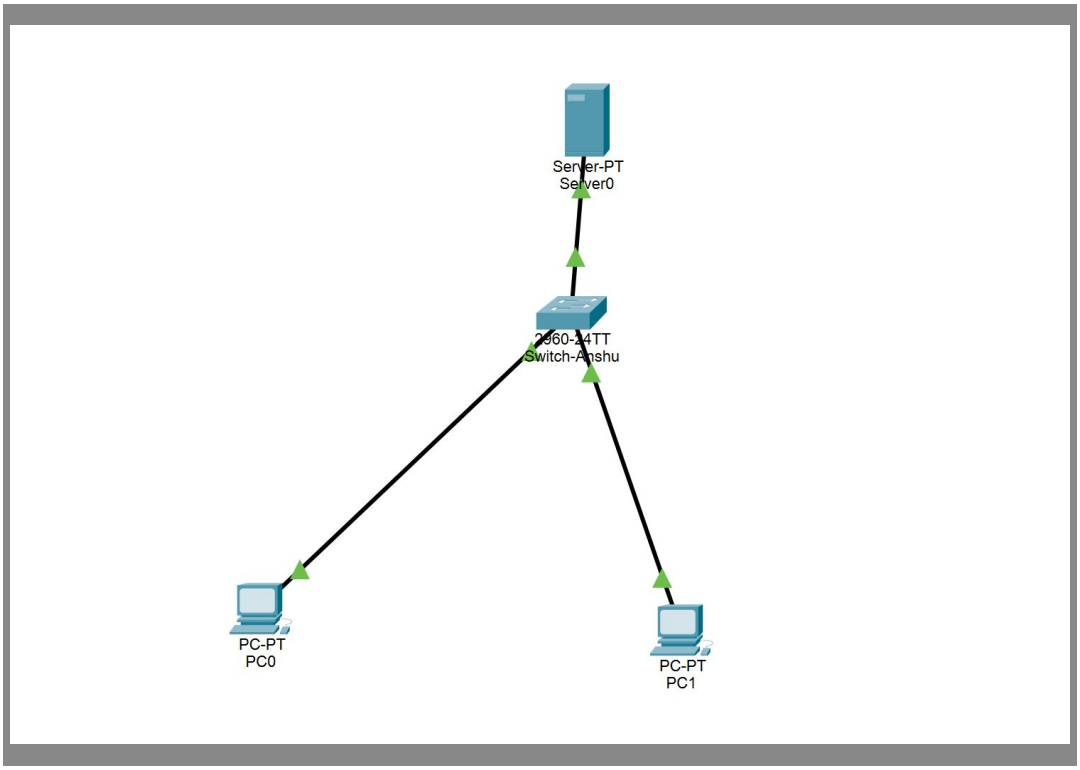
3. Web Server (HTTP)

A Web Server is responsible for storing web content and delivering it to requesting clients.

Operation: It handles communication via the Hypertext Transfer Protocol (HTTP). When a user enters a URL in a browser, the browser sends an HTTP request to the server, which then returns the appropriate HTML content for the browser to render.

Network Topology

A topology was built in Cisco Packet Tracer using one Server (Server-PT named "Server0") connected to a Switch (2960-24TT named "Switch-Anshu"). The Switch connects two client devices, PC0 and PC1. Server0 acts as a multi-purpose device delivering DHCP, DNS, and Web services to all hosts on the network.



IP addressing scheme

IP Configuration Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	DNS Server	Allocation Method
Server0	FastEthernet0	192.168.1.2	255.255.255.0	0.0.0.0	127.0.0.1	Static
PC0	FastEthernet0	192.168.1.3	255.255.255.0	0.0.0.0	192.168.1.2	DHCP
PC1	FastEthernet0	192.168.1.4	255.255.255.0	0.0.0.0	192.168.1.2	DHCP

Configuration:

Server Configuration

Server0 was configured with a static IP address so that clients can reliably locate and connect to it at all times.

- IP Address: 192.168.1.2
- Subnet Mask: 255.255.255.0

DHCP Configuration

The DHCP service was activated on Server0 so that dynamic IP addresses would be automatically distributed to PC0 and PC1.

- Service: On
- Pool Name: LAN_POOL
- Default Gateway: 192.168.1.1
- DNS Server: 192.168.1.2
- Start IP Address: 192.168.1.10
- Maximum number of Users: 50

Web Server and DNS Configuration

1. HTTP Service: Both HTTP and HTTPS services were switched on for Server0. The index.html file was modified to display "Welcome From Anshu."

2. DNS Service: The DNS service was activated on Server0. An address record was created to resolve the domain name to the server's IP address.

Name: www.hcoe.com

Type: A Record

Address: 192.168.1.2

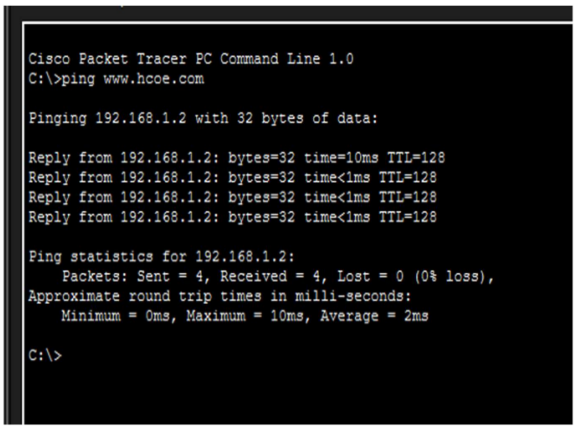
Verifications for all configurations

1. DHCP Verification:

PC0 was opened and its IP configuration was changed to "DHCP". The address request was fulfilled successfully, with IP 192.168.1.3 and the DNS server address both assigned automatically.

2. DNS and Connectivity Verification:

A ping command was executed from PC0 targeting the domain name www.hcoe.com. The DNS server successfully resolved the name to 192.168.1.2, and all four ping replies were received with no packet loss.



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Cisco Packet Tracer PC Command Line 1.0
C:\>ping www.hcoe.com

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=10ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

C:\>
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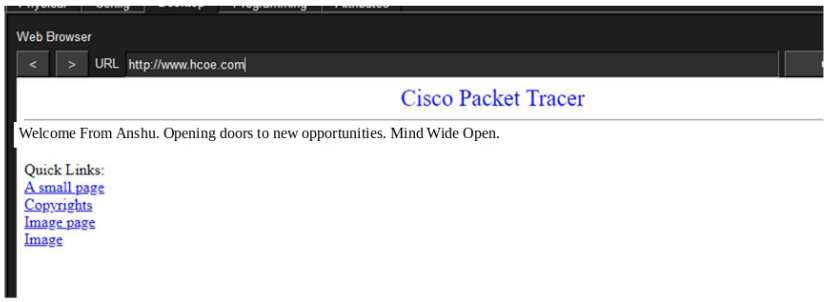
3. Web Server Verification:

A web browser was launched on PC0, and the address http://www.hcoe.com was typed into the URL bar. The customised HTML page stored on Server0 loaded correctly, confirming that the web service was operational.



DISCUSSION AND CONCLUSION

In this lab session, a single server was configured to simultaneously provide DHCP, DNS, and HTTP services to the network. It was observed that both PC0 and PC1 obtained their IP addresses and DNS settings automatically from the DHCP pool on Server0, removing the need for any manual IP configuration on the client machines. The DNS service correctly translated the domain www.hcoe.com into the server's IP address, 192.168.1.2, enabling name-based access. The web page hosted on Server0 was retrieved successfully through the browser using the domain name, verifying end-to-end functionality of all three services.

In summary, the lab was completed successfully, providing a solid understanding and practical exposure to server configuration covering DHCP, DNS, and Web Server services using Cisco Packet Tracer.